



Physiology of nerve and muscles (EXCITABLE TISSUES)

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Objectives

- Define the membrane potential
- Explain the role of ion channels in membrane potential
- Definition and measurements of equilibrium potential
- Define resting membrane potential
- Explain factors that generated the negative RMP



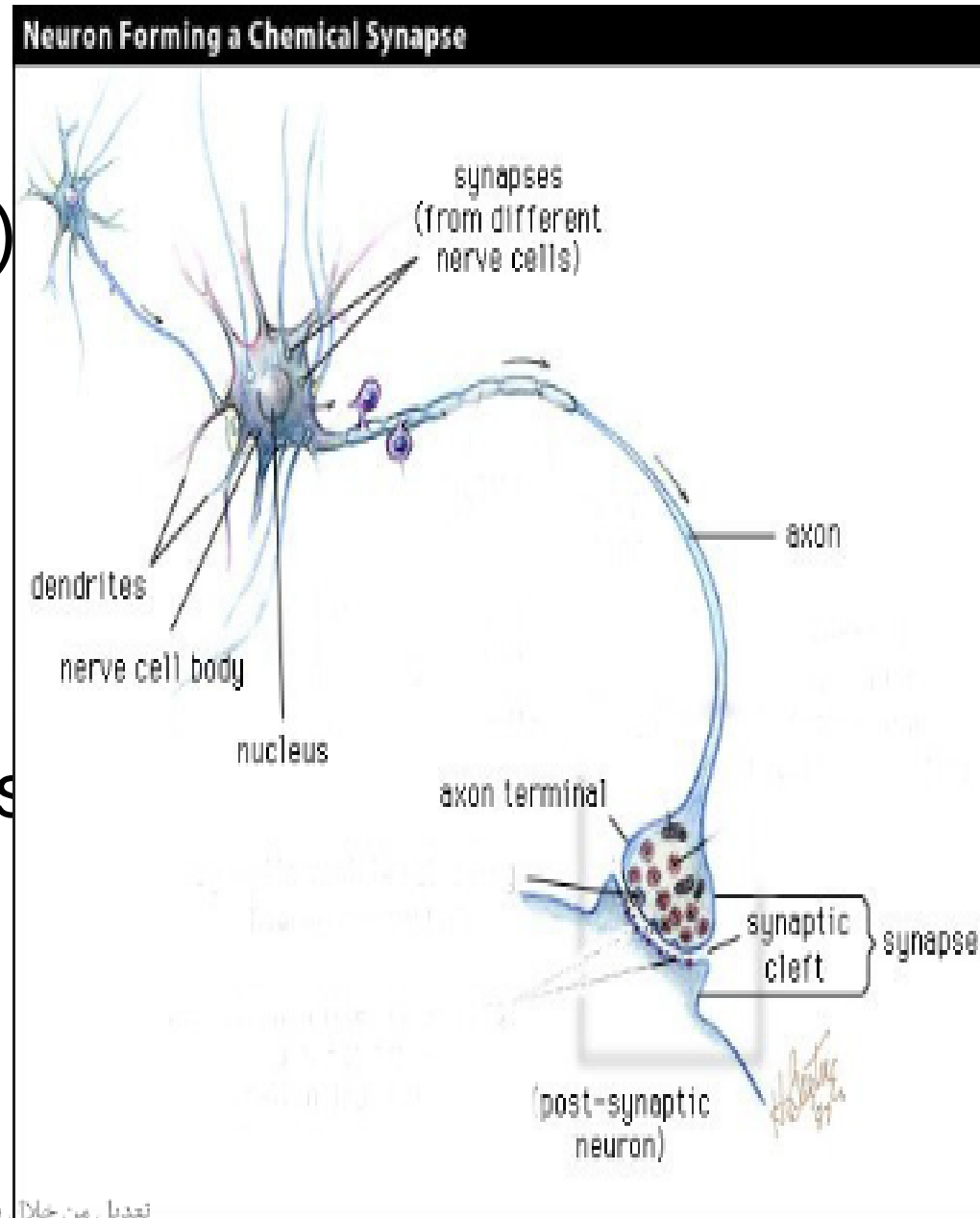
- Neuron or nerve cell is defined as:
the structural and functional unit of nervous system.
 - Neuron is similar to any other cell in the body, having nucleus and all the organelles in cytoplasm.
 - However, it is different from other cells by two ways:
 1. Neuron has branches or processes called **axon** and **dendrites**
 2. Neuron **does not have centrosome.**
- So, it cannot undergo division.

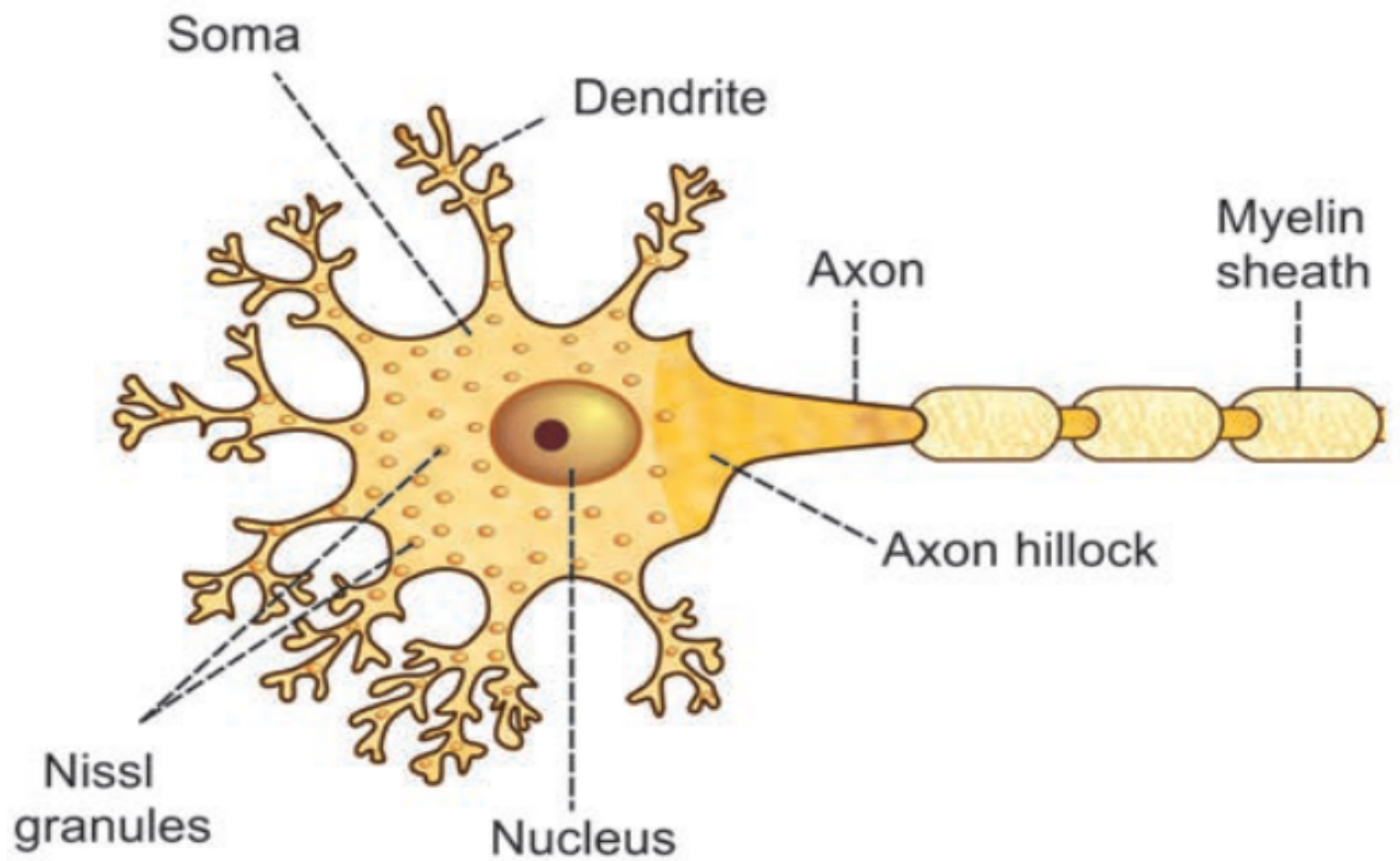


Structural and functional unit of CNS

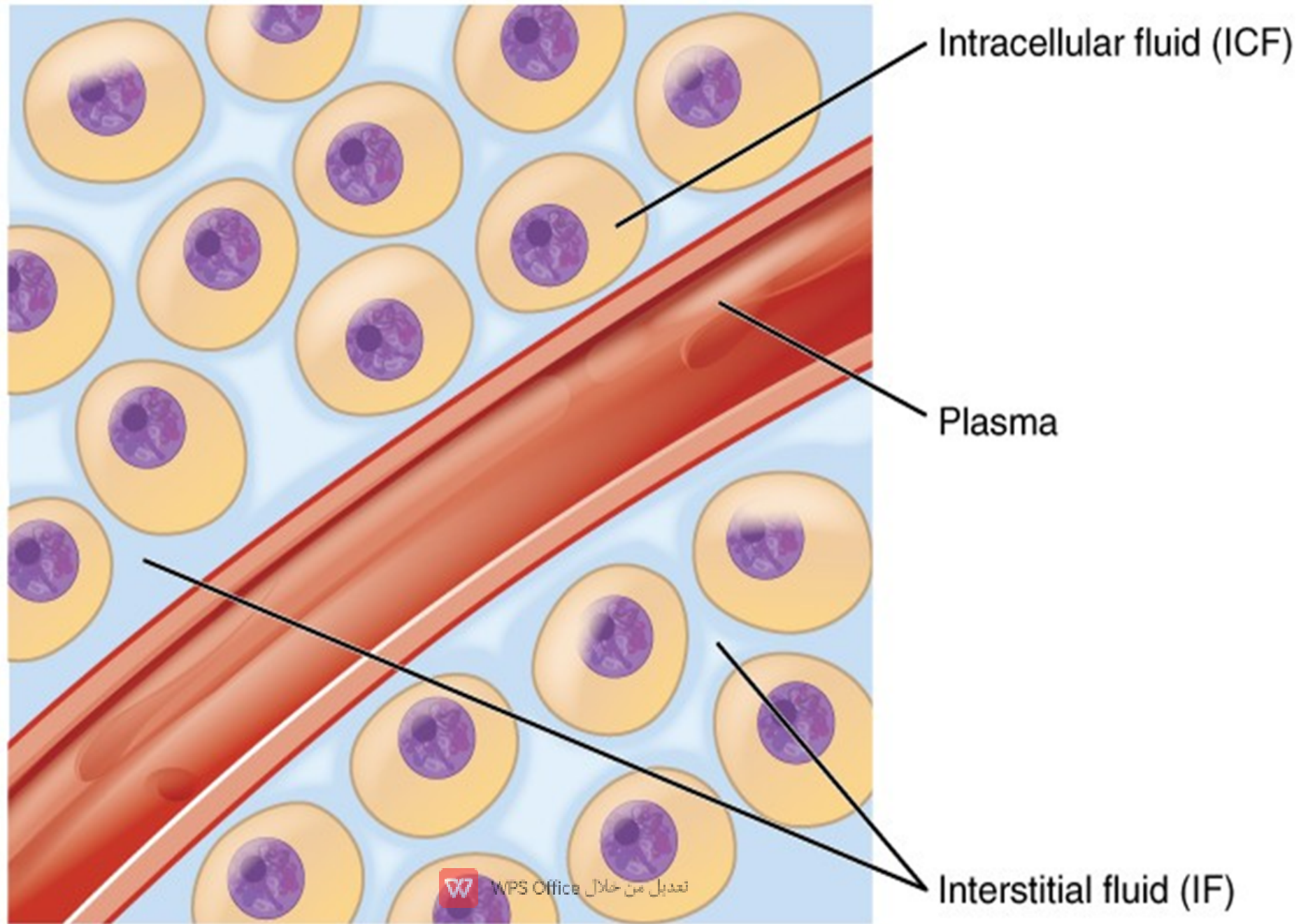
Neuron structure

- **Cell Body (soma)**
- **Dendrites** (input structure) receive inputs from other neurons relay them to the cell body
- **axon** (output structure) a fiber that carries messages (spikes) from the cell to dendrites of other neurons





Human body compartments



Electrolyte composition of Body Fluids

- What is an electrolyte?
- An electrolyte is a substance that when dissolved in water, gives a solution that can conduct electricity.



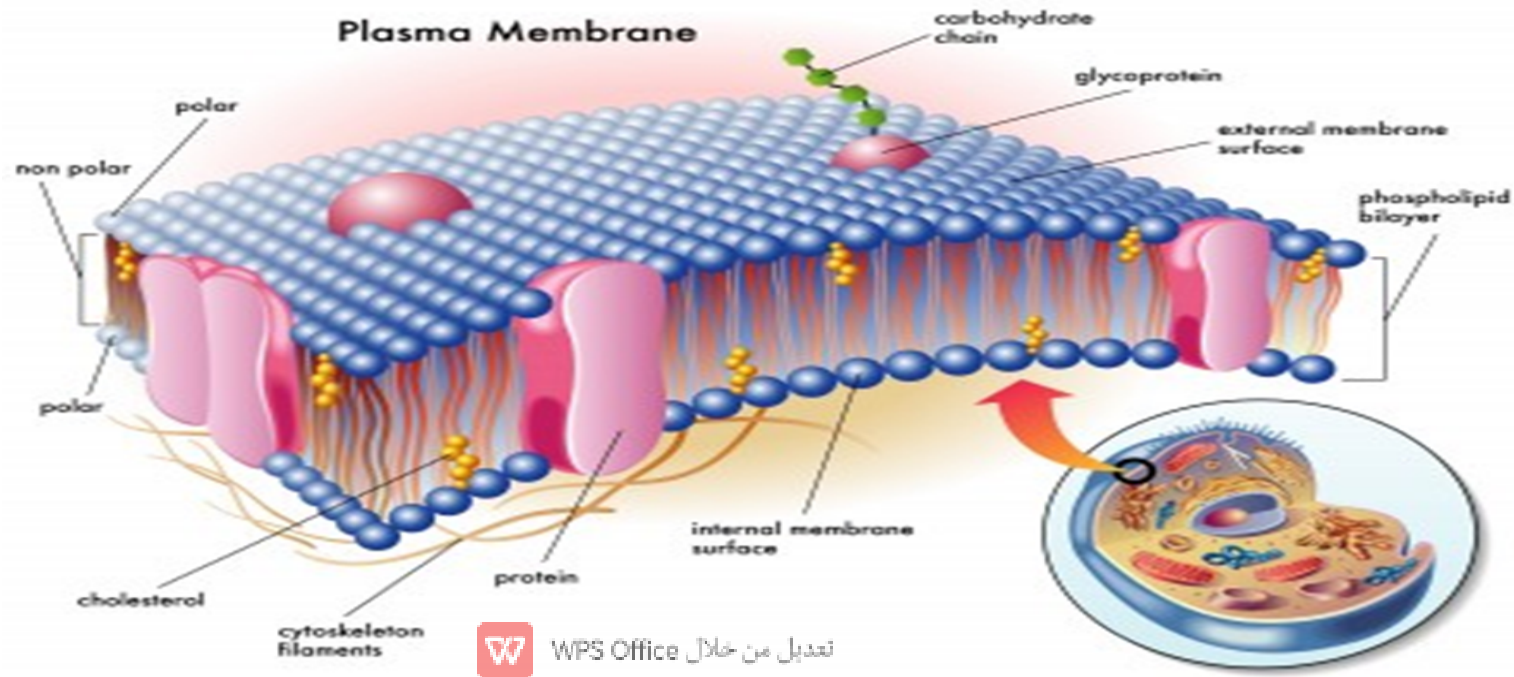
TABLE 6.1: Differences between extracellular fluid (ECF) and intracellular fluid (ICF)

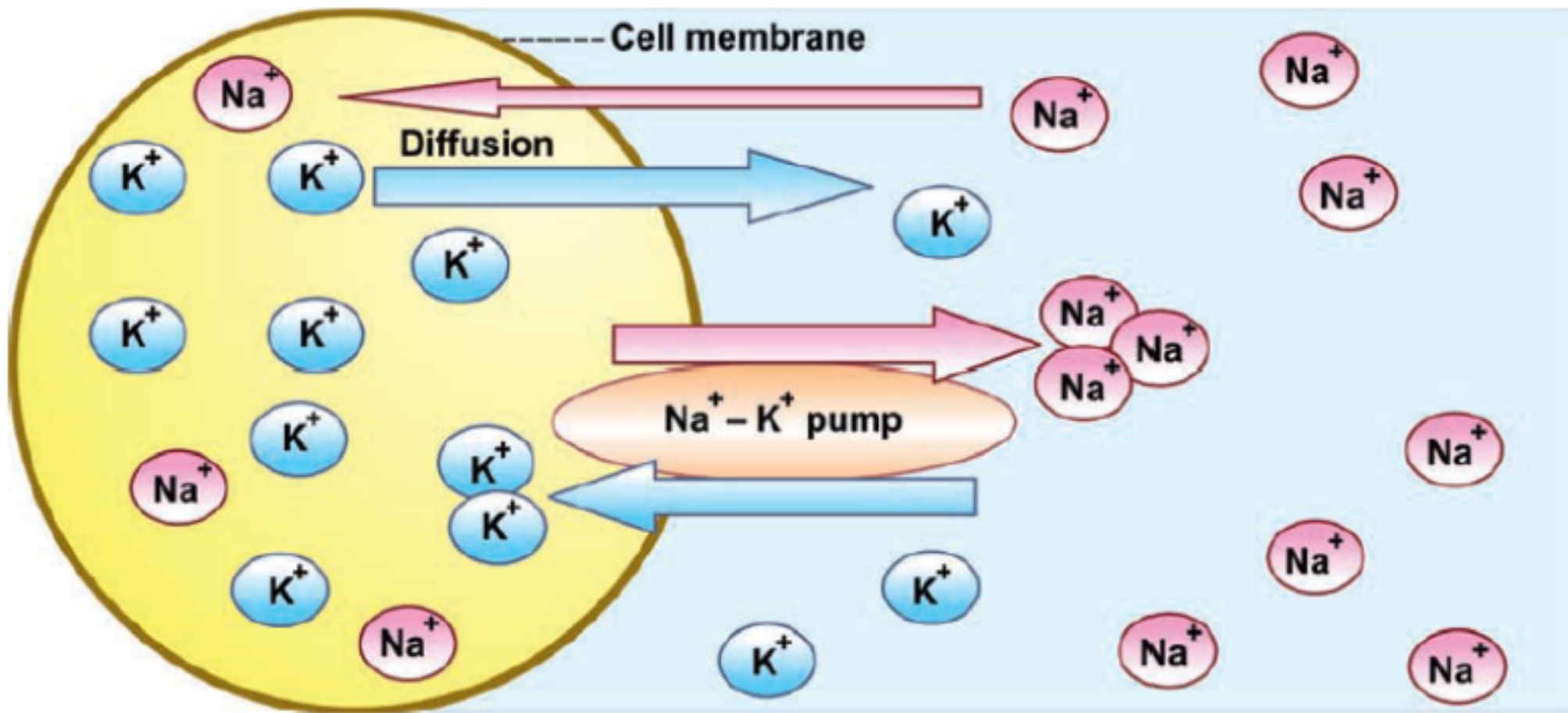
Substance	ECF	ICF
Sodium	142 mEq/L	10 mEq/L
Calcium	5 mEq/L	1 mEq/L
Potassium	4 mEq/L	140 mEq/L
Magnesium	3 mEq/L	28 mEq/L
Chloride	103 mEq/L	4 mEq/L
Bicarbonate	28 mEq/L	10 mEq/L
Phosphate	4 mEq/L	75 mEq/L
Sulfate	1 mEq/L	2 mEq/L
Proteins	2 g/dL	16 g/dL
Amino acids	30 mg/dL	200 mg/dL
Glucose	90 mg/dL	0-20 mg/dL
Lipids	0.5 g/dL	2-95 g/dL
Partial pressure of oxygen	35 mm Hg	20 mm Hg
Partial pressure of carbon dioxide	46 mm Hg	50 mm Hg
Water	15 to 20 L (18)	20 to 25 L (22)
pH	7.4	7.0



Membrane Potential

- It is the electrical potential difference that exist between the inside and the outside of the cell membrane
- It is reported as the inside potential relative to the outside



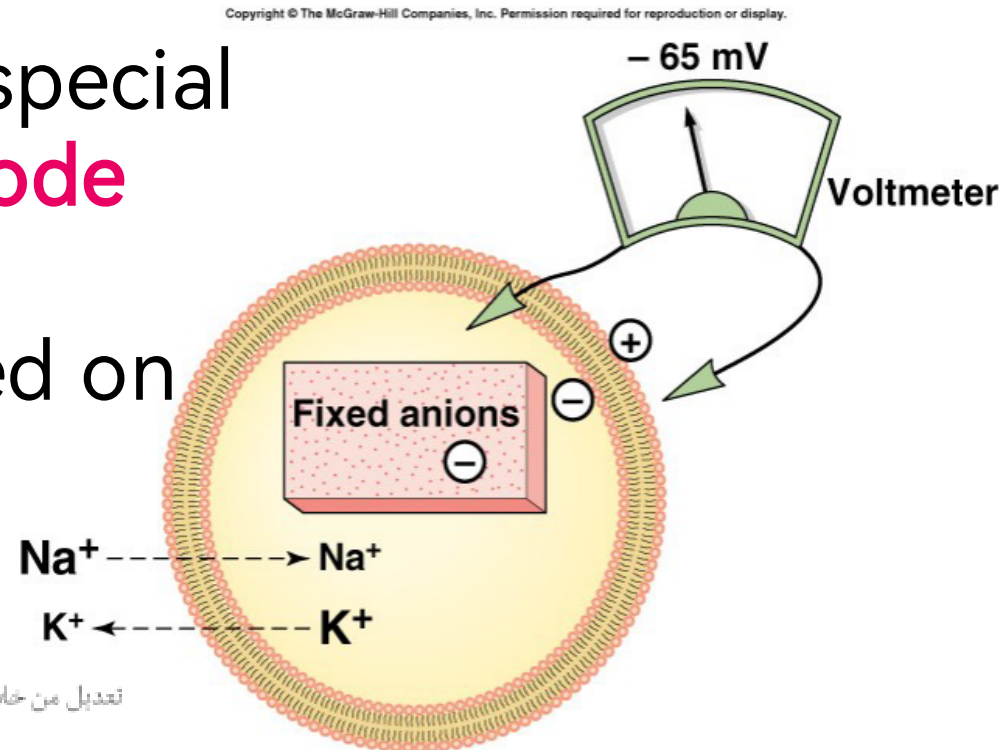


Membrane Potential

- All living cells has a membrane potential (polarized electrically) Due to differences in concentration and permeability of key ions e.g: Na^+ , K^+ and large intracellular proteins between ECF & ICF

*Can be measured by special device called the **Cathode Ray Oscilloscope**

Using electrodes placed on the two sides of the membrane



- Units of electrical potential are in volts (V)
- For biological system the unit is millivolts (mV) ($1\text{ V} = 1000\text{mV}$)
- Voltage is always measured between two points (Potential difference)

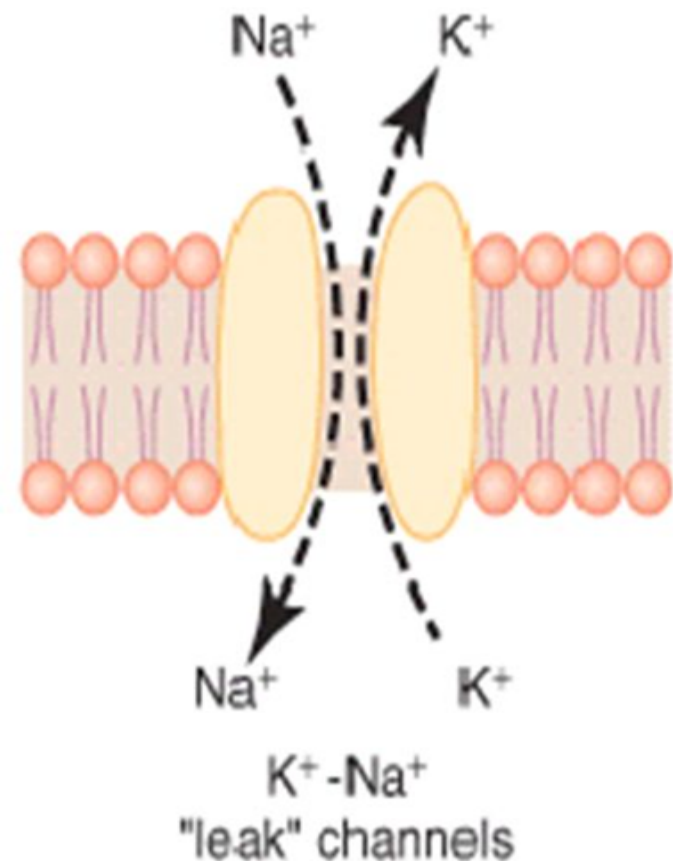


Role of Ion Channels

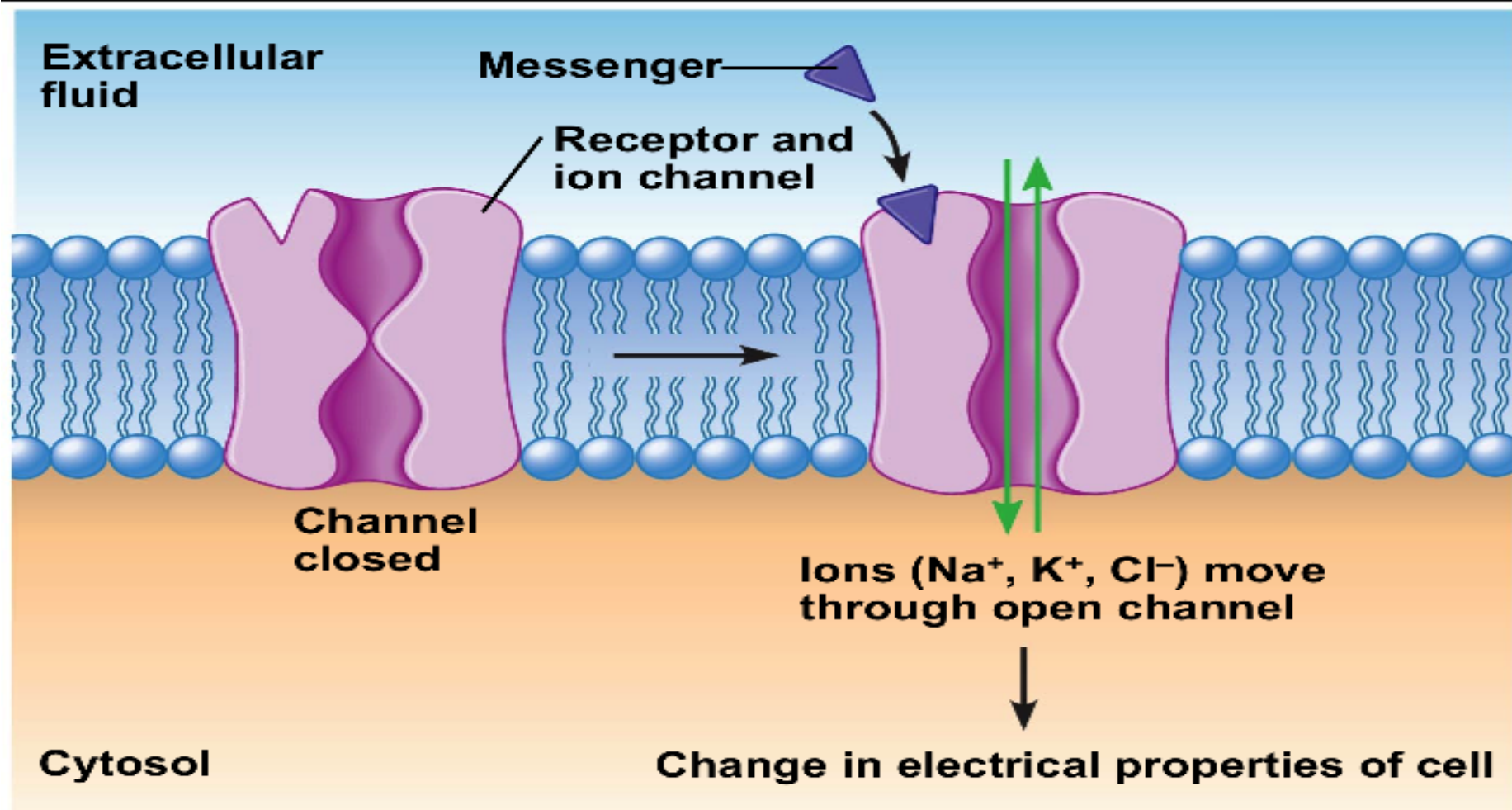
- Passive, or leakage, channels – always open

Role of Na^+ & K^+ leak channels

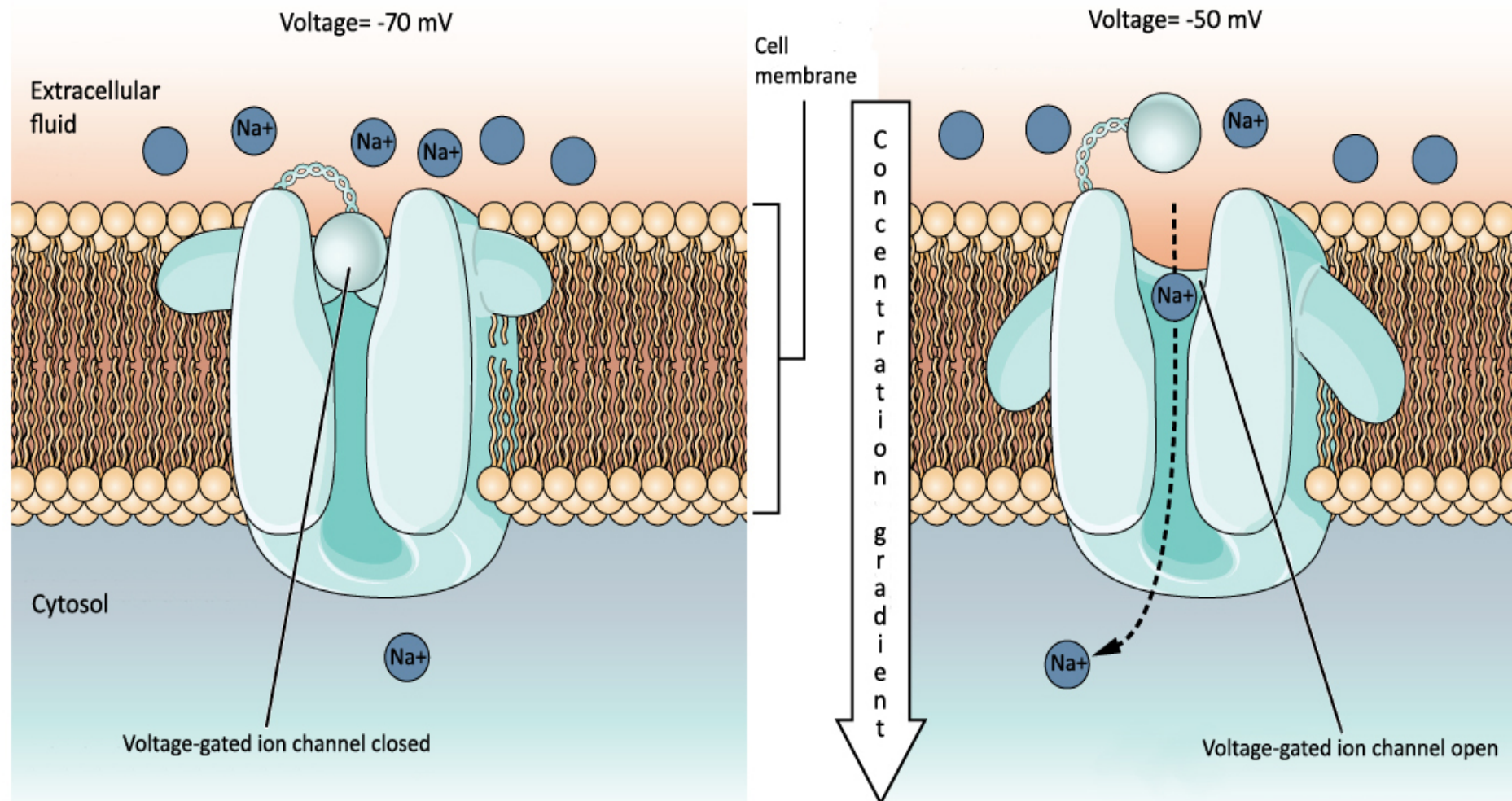
- Exceedingly important
- 100 times more permeable to potassium



- Chemically gated channels – open with binding of a specific neurotransmitter

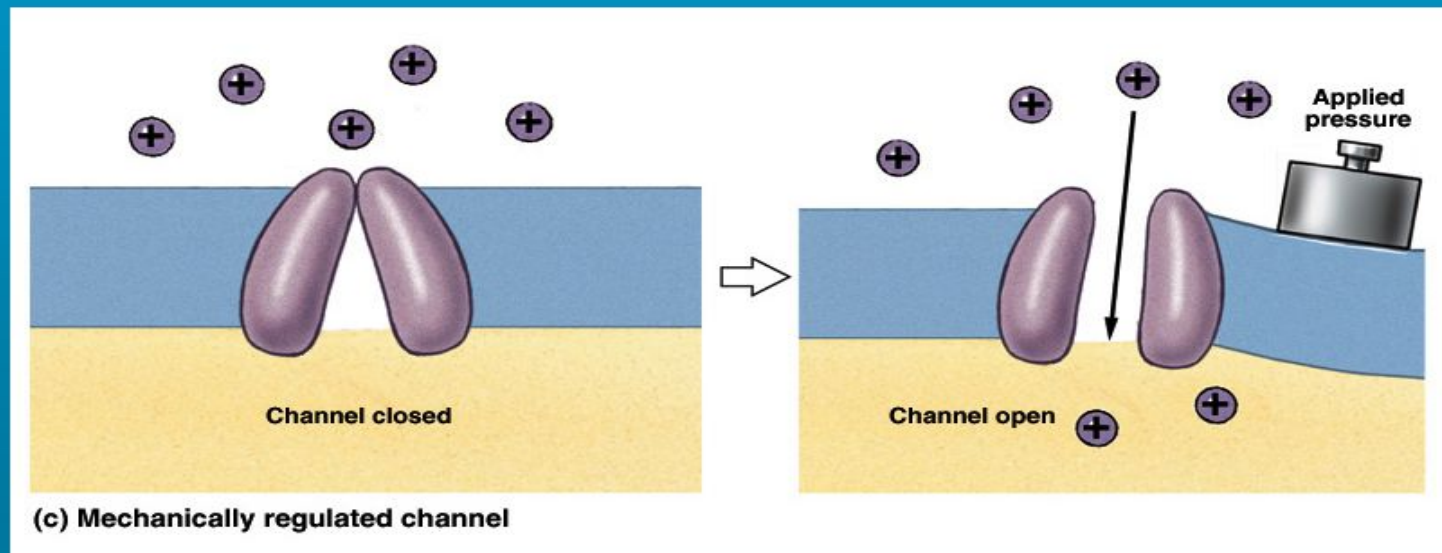


- Voltage-gated channels – open and close in response to change in charge.



- Mechanically gated channels – open and close in response to physical deformation of receptors

Gated channels: mechanically-gated



Equilibrium Potential

- Equilibrium potential at which ion movements in both directions across the membrane are exactly balanced .
- The value of the equilibrium potential (**Nernst potential**) for any ion depends on the concentration gradient across the membrane for that ion



- When the concentration gradient of ion is given, the Nernst potential for this ion can be calculated by the Nernst equation



The Nernst equation

$$E_{\text{ion}} = 61 \log [ion]_o / [ion]_i$$

- E_{ion} = equilibrium potential for ion in mV
- $[ion]_o$ = concentration of the ion outside the cell
- $[ion]_i$ = concentration of the ion inside the cell



Goldman Equation

- When a membrane is permeable to several different ions

$$E_m = \frac{RT}{F} \ln \frac{P_K[K^+]_o + P_{Na}[Na^+]_o + P_{Cl}[Cl^-]_i}{P_K[K^+]_i + P_{Na}[Na^+]_i + P_{Cl}[Cl^-]_o}$$

Diagram illustrating the Goldman Equation, showing the relationship between membrane potential (E_m) and ion concentrations and permeabilities.

Labels and arrows indicating the components of the equation:

- Gas constant (R)
- Temperature (T)
- Faraday constant (F)
- Permeability of Potassium (P_K)
- Concentration of potassium outside the cell ($[K^+]_o$)
- Concentration of potassium inside the cell ($[K^+]_i$)

- Different types of Membrane potentials:
- Resting membrane potential
- Action potential
- Local potential
- Neuromuscular junction



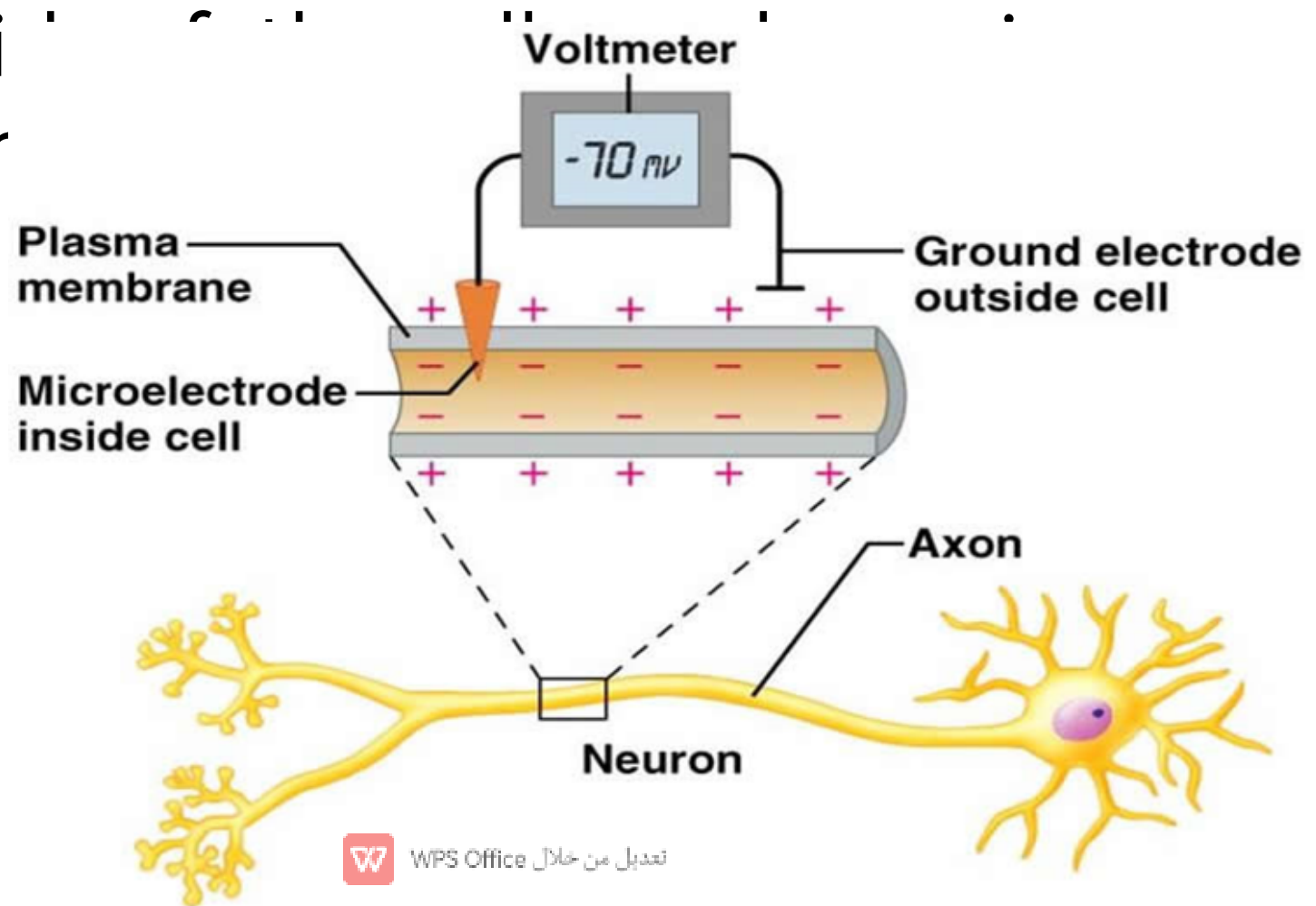


RESTING MEMBRANE POTENTIAL



- The Resting membrane potential (RMP):

It is the constant potential difference that exist between the inside and the outside of a cell at rest.



- All living cells have RMP
- The RMP is always negative
- I.e. the inside of the membrane is more negative than the outside



- The magnitude of the RMP (-50 to -90), differs in different types of cells, for example :
- In a nerve cell = -70 mV
- In skeletal muscle cell = -90 mV (i.e. the potential inside the fiber is 90 millivolts more negative than the potential in the ECF)



- What are the factors that generated the negative RMP ?

1-The different concentration of ions (Na⁺ and K⁺) between the ICF and the ECF:

K⁺ is the main ICF cation and will diffuse out off the cell following the concentration gradient

When K⁺ diffuse out of the cell, the inside of the cell loses a positive charge (become more negative)



2-Permeability of the cell membrane to K^+ is more than its permeability to Na^+ during rest:

- The leak channels are far more permeable to K^+ than to Na^+ (100 times more) that they are called K^+ leak channels or K^+ channels
- Much more K^+ ions can diffuse out off the cell (making the inside negative) than Na^+ ions diffusing in (making the inside positive)



3- The action of Na⁺/K⁺ ATPase pump:

- Every time the Na⁺,K⁺ ATPase pump transport 3 positive charges outside of the cell, they are replaced by only 2 positive charges = the inside becomes more negative



4-The effect of intracellular nondiffusible anions (proteins and phosphates):

- Because proteins and phosphates are negatively charged and can not diffuse out of the membrane they affect the distribution of permeant ions (Gibbs-Donnan effect)



- Which of these factors is more dominant in determining the value of the normal RMP ?

The permeability of the cell membrane to K^+



